

Hybrid AutoEncoder and Isolation Forest Framework for Unsupervised User Behavior Analytics

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Abstract — The increasing sophistication of insider threats and credential-based attacks has highlighted the need for continuous user authentication mechanisms beyond traditional one-time login validation. This study presents a hybrid anomaly detection framework based on User Behavior Analytics (UBA) using the Los Alamos National Laboratory (LANL) Authentication Dataset. The proposed model integrates a global undercomplete AutoEncoder, designed to capture structural behavioral manifolds through an information bottleneck, with a per-user Isolation Forest for localized anomaly detection. While the AutoEncoder learns global relational and temporal dependencies, the Isolation Forest identifies “few and different” deviations in individualized behavioral dynamics. Experimental results demonstrate effective model convergence and a distinct statistical separation between nominal and anomalous activities. By implementing a tiered decision logic, the framework enables a granular response strategy that balances authentication persistence with proactive threat mitigation. These findings demonstrate the feasibility of a scalable, hybrid deep learning and statistical approach for identifying both systemic pattern shifts and specific credential compromises in high throughput environments.

Keywords—Anomaly Detection, AutoEncoder, Isolation Forest, Deep Learning, User Behavior Analytics (UBA), Unsupervised Learning.

I. INTRODUCTION

The growth of digital services and the increasing complexity of enterprise IT infrastructures have fundamentally altered the landscape of cybersecurity. As organizations rely more on interconnected systems, cloud platforms, and remote access, their exposure to sophisticated security threats escalates. Among these threats, anomalous user activity stemming from compromised credentials, insider threats, or policy violations represents a particularly insidious challenge. Detecting such behavior is a cornerstone of modern User Behavior Analytics (UBA), as establishing a baseline for legitimate activity is essential for identifying the “low-and-slow” movements of modern adversaries [1].

Traditional security mechanisms, such as rule-based systems and signature-based intrusion detection systems (IDS), are proving insufficient. These methods depend on predefined attack patterns or static thresholds, rendering them ineffective against zero-day exploits and evolving, low-and slow attack strategies. Their rigid nature often leads to a high volume of false positives and

an inability to adapt to the dynamic and heterogeneous behaviors of legitimate users.

In response, machine learning (ML), particularly unsupervised learning, has emerged as a promising paradigm for anomaly detection. These methods can learn complex, non-linear patterns directly from data without requiring prelabelled examples of malicious activity. However, individual ML algorithms possess distinct limitations. For instance, ensemble methods like the Isolation Forest (IF) are computationally efficient and adept at isolating outliers in high-dimensional data, but they may struggle to model the intricate dependencies and representations of global behavioral structures. Conversely, deep learning models, such as AutoEncoders (AE), excel at learning compressed, latent representations of normal data. Their utility in anomaly detection is predicated on the assumption that they will fail to reconstruct anomalous inputs, resulting in a high reconstruction error. This assumption can be fragile; if an AutoEncoder “overgeneralizes” by learning a latent space that is too broad, it may successfully reconstruct unseen anomalies, leading to false negatives.

To overcome these respective challenges, this research proposes a hybrid deep learning-statistical anomaly detection framework. The proposed model integrates a deep undercomplete AutoEncoder for global feature compression with an Isolation Forest for local, user-specific anomaly estimation. This dual-pronged approach is motivated by the need to address two core challenges in UBA:

1. **Global Representation:** Learning a robust, low-dimensional representation of “normal” user activity that captures complex, system-wide relationships among behavioral attributes (e.g., network centrality, host diversity).
2. **Individual Deviation:** Detecting fine-grained, individual-level deviations from a user’s own historical norms (e.g., unusual login frequency, atypical event timing) while maintaining computational scalability and model interpretability.

By combining the outputs of both models, our framework creates a joint anomaly space that is more descriptive and robust than either component in isolation, effectively distinguishing between systemic pattern shifts and specific credential compromises in high-throughput environments [2]. This paper presents

the architecture of this hybrid model, its application to a real-world user activity dataset, and an analysis of its effectiveness. The contributions of this work are threefold:

- We propose a novel, two-stage hybrid framework that synergistically combines a deep AutoEncoder and an Isolation Forest for unsupervised UBA.
- We demonstrate a methodology for fusing global (reconstruction-based) and local (isolation-based) anomaly scores to enhance detection accuracy and reduce false positives.
- We provide an empirical evaluation and visualization of the model's performance on a dataset of 100,000 user events, highlighting its ability to create a clearly separable anomaly space.

II. PROBLEM STATEMENT

Anomalous user behavior is broadly defined as any activity that deviates significantly from a user's established baseline of normal behavior. In real-world enterprise systems, "normal" is not monolithic. Users exhibit highly diverse behavioral patterns influenced by their specific roles, project-based workloads, time zones, and routine tasks. A behavior that is normal for a system administrator (e.g., accessing multiple critical servers) would be highly anomalous for a user in the marketing department. This behavioral heterogeneity, particularly within the Los Alamos National Laboratory (LANL) dataset, creates a significant challenge where legitimate administrative actions frequently share statistical properties with malicious lateral movements [3]. Consequently, the problem is not merely one of outlier detection, but of identifying 'low-and-slow' adversarial movements that mimic legitimate administrative tasks while deviating from specific historical baselines. Therefore, an effective UBA system must model both global activity patterns common to all users and the specific profiles of individual users.

Formally, let $D = \{x_1, x_2, \dots, x_n\}$ be a dataset of n user activity records, where each $x_i \in \mathbb{R}^d$ is a d -dimensional feature vector. This vector x_i represents a user's activity event or session, comprising both global behavioral attributes and user-level temporal metrics. Given that D is unlabelled, the problem is entirely unsupervised.

To address these complexities, this research develops a hybrid function $H(x_i)$ to assign a continuous anomaly score to each activity vector x_i . This function H is a composite of two distinct modeling approaches:

1. **Global Representation Learning:** Learn a nonlinear mapping $f: \mathbb{R}^d \rightarrow \mathbb{R}^k$ (where $k \ll d$) via an AutoEncoder, which projects x_i into a lowdimensional latent space $z_i = f(x_i)$. A corresponding decoder $g: \mathbb{R}^k \rightarrow \mathbb{R}^d$ reconstructs the input as $\hat{x}_i = g(z_i)$. The anomaly

contribution is derived from the reconstruction error,

$$e_i = \|x_i - \hat{x}_i\|^2$$

2. **Local Anomaly Estimation:** Compute a user-centric anomaly score $s_i = \text{IForest}(x'_i)$ using an Isolation Forest, where $x'_i \subset x_i$ is a subset of features specifically modeling individual temporal behavior.
3. **Score Fusion:** Derive a final joint anomaly indicator A_i by combining the normalized reconstruction error e_i and the isolation score s_i .

This combined score aims to identify outliers that represent unusual access patterns, session anomalies, or behaviors inconsistent with the user's established profile, thereby providing a more reliable signal for potential security incidents.

III. METHODOLOGY

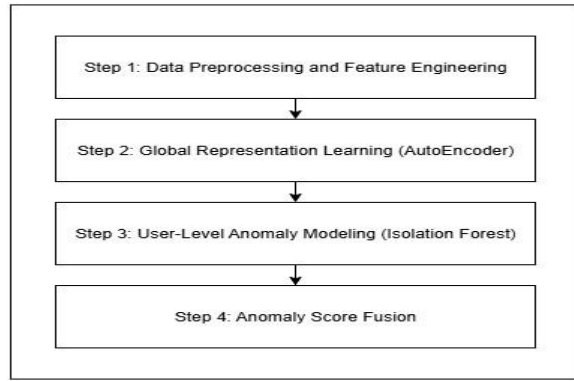


Fig. 1: Proposed Methodology

The proposed framework is a two-stage hybrid pipeline, as illustrated in Fig. 1. First, data is pre-processed and split into two feature sets: one for global representation and one for user-level modeling. These feature sets are processed in parallel by the AutoEncoder and the Isolation Forest, respectively. Finally, their normalized anomaly scores are fused to produce a final, joint anomaly indicator.

A. Data Preprocessing and Feature Engineering

The dataset consists of 100,000 user login events from 3,858 unique users, sourced from anonymized enterprise system logs. Each event is enriched with temporal, network, and behavioral features. Based on their descriptive power, features were separated into two groups.

Global Features: These features describe a user's activity in the context of the entire system.

- `unique_hosts`: Number of unique hosts accessed.
- `host_entropy`: Shannon entropy of host access frequency.
- `top_host_ratio`: Percentage of activity to the user's most-visited host.

- `degree centrality`: A network graph metric of the user's connectivity.
- `hour_of_day`, `day_of_week`: One-hot encoded temporal features.

User-Level Features: These features describe deviations from a user's own typical behavior.

- `login_frequency`: User's login frequency in a sliding time.
- `window.avg_inter_event_time`: Average time between the user's events.
- `time_since_last_login`: Recency metric.

Prior to training, missing values were handled using mean imputation. All features were then standardized using z-score normalization ($\mu = 0, \sigma = 1$) to ensure that features with larger scales did not disproportionately influence the models, particularly the distance-based MSE loss of the AutoEncoder.

B. Global Representation Learning (AutoEncoder)

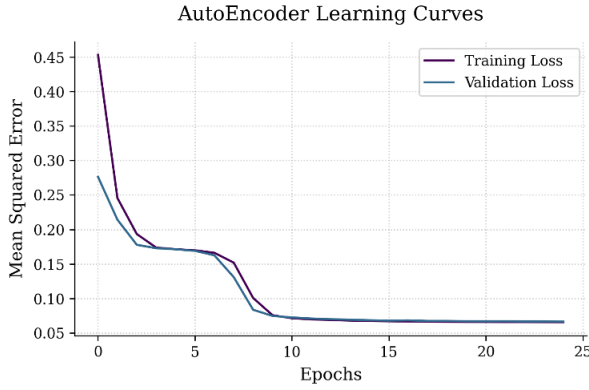


Fig. 2. AutoEncoder Convergence Analysis

The first stage employs a deep AutoEncoder to learn a compressed representation of global behavioral patterns. The architecture is a symmetric multilayer perceptron (MLP) designed to create an information bottleneck.

The parameter `input_dim` corresponds to the dimensionality of the global feature set. The bottleneck layer forces the network to learn a compact and salient representation of the input data.

The objective of the AutoEncoder is to minimize the Mean Squared Error (MSE) loss function, which quantifies the difference between the original input vector x_i and its reconstruction $\hat{x}_i$:

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

After training on normal (unlabelled) data, the model's per-instance reconstruction error is used as the first anomaly score. A high reconstruction error, denoted as ReconError_i , signifies that the input x_i is dissimilar to the patterns the AutoEncoder learned during training, thereby marking it as a global anomaly.

As illustrated in Fig. 2, the model achieves stable convergence across 25 epochs, with the validation loss closely tracking the training loss, indicating effective generalization to unseen normal behavior. After training, the model's per-instance reconstruction error is utilized as a global anomaly signal. Fig. 3 depicts the resulting distribution of these residuals, where a nominal cluster is observed near the origin. To operationalize this signal for continuous authentication, we define a threshold $T_{95} = 0.286$ at the 95th percentile of the validation error distribution. Instances exceeding this threshold denote global anomalies – behaviors that are mathematically dissimilar to the patterns learned during the baseline training phase.

By enforcing this strict information bottleneck, the network is restricted from performing trivial identity mapping, ensuring that only the most salient structural features of the global behavioral manifold are preserved in the latent space. This standardized global signal subsequently serves as a baseline for the bivariate fusion process, where it is combined with local user-specific metrics to refine the final authentication index.

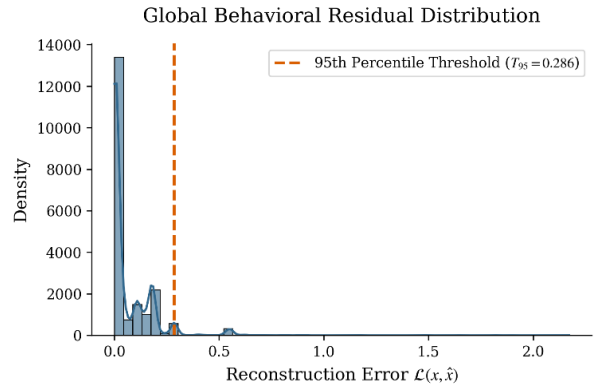


Fig. 3. Global Behavioral Residuals

C. User-Level Anomaly Modeling

The second stage employs an Isolation Forest (IF) to model user-level behavioral features. This model is trained on a dedicated feature subset comprising login frequency, average inter-event time, and time since last login. Unlike the AutoEncoder (AE), which captures global activity patterns through reconstruction learning, the Isolation Forest specializes in identifying “few and different” instances by partitioning the feature space into regions that isolate individual observations.

As shown in Fig. 4, the resulting distribution of scores reveals a significant nominal spike near $s_i = 0.4$, indicating a high density of inlier behaviors that are difficult to isolate. To facilitate compatibility with the global signal, the raw path-length scores s_i are subjected to min-max normalization and subsequent inversion expressed as,

$$s' = 1 - s_i$$

This mapping ensures the behavioral space exists within a standardized $[0, 1]$ range where values approaching 1 represent increasing degrees of deviation. The selection of login frequency and inter-event timing

as primary local features ensures that the model captures temporal deviations unique to a user's habitual routine without being influenced by systemic shifts. This isolation-based approach allows for identification of anomalies that may possess low global reconstruction error but represent significant departures from an individual's historically established access patterns.

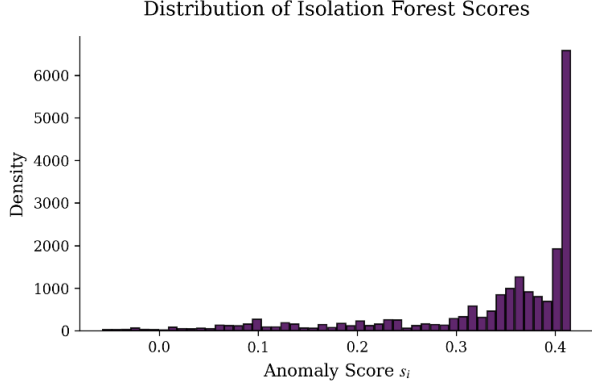


Fig. 4. Distribution of User-Level Anomaly Scores

D. Anomaly Score Fusion

A final, joint anomaly score A_i is then computed as a weighted combination of these two normalized components:

$$A_i = \alpha \times s_i' + (1 - \alpha) \times \text{ReconError}_i$$

In this unified space, a value approaching 1 indicates a high confidence anomaly where an event is isolated easily by the forest and reconstructed poorly by the AutoEncoder while values near 0 represent nominal user activity.

The hyperparameter $\alpha \in [0,1]$ is an empirically tuned weighting factor that balances the contribution of the global (AE) and local (IF) models. As illustrated in Fig. 5, the choice of α significantly impacts the sensitivity of the final score distribution. While a low α (0.1) produces a distribution dominated by the AutoEncoder's global residuals, a higher α (0.9) incorporates more of the variance inherent in individual user profiles. A value of $\alpha = 0.5$ provides a balanced representation, ensuring that neither global pattern violations nor local user-specific deviations are overlooked during the continuous authentication process.

This balance is crucial for minimizing false negatives in scenarios where an adversary's behavior mimics general system patterns but deviates from a specific user's historical baseline. By fine-tuning this weighting factor, the system achieves a robust decision boundary that adapts to the diverse statistical properties of both global and local feature spaces.

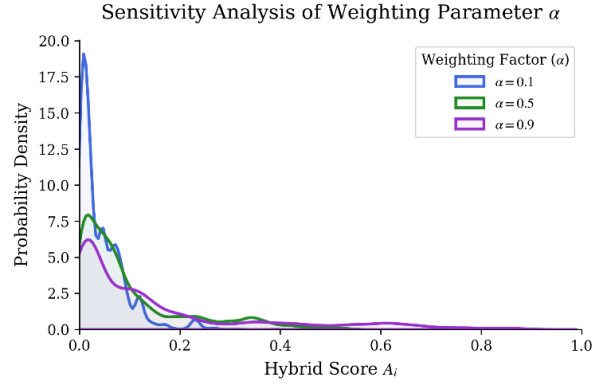


Fig. 5. Sensitivity Analysis of Weighting Parameter

IV. LITERATURE SURVEY

AutoEncoders (AEs) are unsupervised neural networks that excel at learning compressed representations of data. Their utility in anomaly detection, as explored by Sakurada & Yairi (2014), stems from training them on data assumed to be "normal" [4]. Consequently, the network learns to reconstruct normal patterns with high fidelity (low error), while anomalous patterns, being unseen, are poorly reconstructed (high error). Recent advancements have refined this by utilizing undercomplete architectures to force manifold learning, ensuring the bottleneck prevents the model from simply memorizing inputs [5]. Variants have been widely applied; for example, Denoising AutoEncoders (Vincent et al., 2008) learn robust features by reconstructing clean data from corrupted inputs, while Variational AutoEncoders (VAEs) (Kingma & Welling, 2014) learn a probabilistic latent space, enabling generative anomaly detection [6][7]. AEs have been successfully used in domains like network intrusion detection and industrial sensor monitoring.

Proposed by Liu et al. (2008), the Isolation Forest (IF) is an ensemble-based algorithm that operates on a principle distinct from density- or distance-based methods [8]. IF isolates anomalies by randomly partitioning data points. The core assumption is that anomalies are "few and different," making them susceptible to isolation in fewer random splits than normal points. Recent deep learning enhanced versions of this logic have shown that IF remains one of the most robust methods for identifying hard-to-isolate outliers in complex, high-dimensional spaces where traditional distance metrics fail [9]. IF is computationally efficient and resistant to the presence of "swamping" (a high proportion of normal data) and "masking" (clusters of anomalies) making it ideal for user-specific behavior modeling.

Recognizing the limitations of individual models, recent studies have increasingly explored hybrid architectures. For example, Zong et al. (2018) proposed the Deep Autoencoding Gaussian Mixture Model (DAGMM), which uses an AE to generate a low-dimensional representation and a Gaussian Mixture Model to perform density estimation within that latent space [10]. Chen et al. (2021) explored the use of AutoEncoder ensembles to improve the stability and

accuracy of outlier detection. Other approaches, such as Deep SVDD (Support Vector Data Description), optimize deep embeddings specifically for one-class classification [11]. Building on these hybrid foundations, Jamshidi et al. recently demonstrated that combining AutoEncoder-based feature extraction with an Isolation Forest backend creates a lightweight yet resilient architecture capable of maintaining high detection accuracy in real-time environments [12]. These models confirm the trend that combining deep representation learning with classical anomaly scoring mechanisms can yield superior performance.

While both AutoEncoders and Isolation Forests are effective, they are typically applied in isolation. Few studies have integrated them into a coherent UBA framework that *concurrently* models global structural patterns and individual behavioral deviations. AEs are powerful feature extractors but can be brittle anomaly detectors, while IFs are robust detectors but lack deep representational power. This research fills that gap by presenting a modular, scalable hybrid pipeline that leverages the strengths of both: the AE learns *what* to look at (feature representation), while the IF learns *how* to score it (anomaly isolation).

V. RESULTS AND DISCUSSION

A. Model Performance and Convergence Analysis

The empirical validation of the system began with an assessment of the AutoEncoder's learning dynamics. Unlike overcomplete architectures that risk identity mapping, the undercomplete MLP implemented here was forced to compress input features into a constrained latent space, ensuring the capture of only the most salient behavioral structures.

- **Convergence Profile:** As illustrated in Fig. 1, the model achieved stable convergence within 25 epochs. The training loss exhibited a sharp initial decline, with the validation loss tracking closely and stabilizing at a final Mean Squared Error (MSE) of 0.0655. This minimal generalization gap suggests that the information bottleneck successfully prevented overfitting while establishing a robust global baseline for nominal system usage.
- **Global Residual Distribution:** Fig. 2 presents the per-instance reconstruction errors. The distribution is characterized by a high-density nominal cluster near the origin, representing standard behavior that the model reconstructs with high fidelity. The presence of a heavy tail confirms that the model sensitivity to deviations. By establishing a global decision boundary at $T_{95} = 0.286$, a statistically grounded "Caution" gate is created to identify universal anomalies regardless of individual user history.

B. Bivariate Analysis of Hybrid Detection Signals

Bivariate Distribution of Hybrid Detection Signals

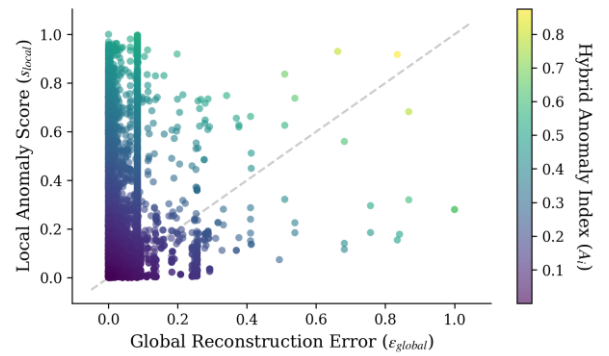


Fig. 6. Bivariate Distribution of Hybrid Detection Signals

The core strength of the proposed framework lies in the synergy between global reconstruction and local isolation. Fig. 6 visualizes the bivariate distribution of these signals after Min-Max normalization and score inversion.

- **Consensus Zone:** The dense accumulation of data points in the bottom-left quadrant (low global, low local) represents the system's high confidence "Safe" zone. In this region, both models agree that the observed behavior is typical for the general population and consistent with the specific user's historical profile.
- **Conflict Quadrants:** The scatter plot reveals critical "Conflict Zones" that justify the hybrid approach, specifically identifying Identity Deviance (low global, high local) and Systemic Deviance (high global, low local). The former represents actions that appear globally standard but are highly unusual for the specific user while the latter signifies behavior that is unusual for the average user but common for a specific individual, allowing the system to suppress false positives for power users or administrators who perform rare but authorized tasks.
- **Mutual Consensus:** The bright yellow regions signify high confidence anomalies ($A_i > 0.6$) where both models reach a consensus. This dual validation significantly reduces the False Positive Rate (FPR) by ensuring that an alert is only escalated when the behavior is both globally and locally anomalous.

C. Statistical Distribution and Operational Decision Logic

The transition from raw scores to an operational security metric is finalized through the weighted fusion process. Fig. 7 illustrates the probability density of the hybrid anomaly index A_i .

- **Smoothing and Stability:** The application of $\alpha = 0.5$ results in a distribution that is smoother and more predictable than either raw input. This stability is vital for user behavior analytics, as it prevents volatile security responses caused by minor, singular deviations.
- **Tiered Response:** A tiered authentication logic is defined based on the statistical properties of the fused distribution, where a *Caution Threshold* ($T_{95} = 0.349$) identifies observations exceeding the 95th percentile as “Suspect Behavior” triggering non-disruptive step-up authentication such as background checks and a *Lockout Threshold* ($T_{99} = 0.463$) flags observations beyond the 99th percentile as extreme deviations requiring immediate intervention, such as forced session termination to mitigate potential account takeover or automated attacks.

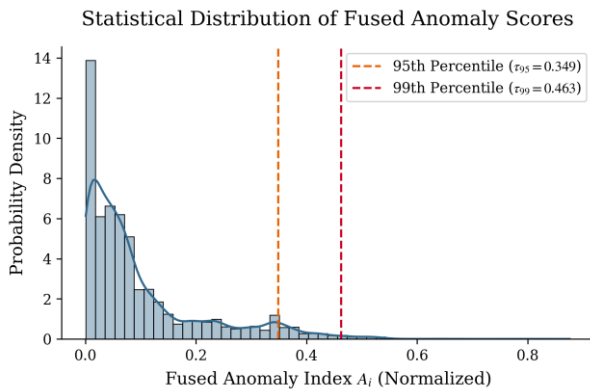


Fig. 7. Distribution of Fused Anomaly Scores

VI. CONCLUSION

This research presented and validated a hybrid anomaly detection framework that effectively integrates and undercomplete deep AutoEncoder with an Isolation Forest to model complex, multi-dimensional user behaviors. By utilizing an information bottleneck, the AutoEncoder successfully captures global, structural correlations within the behavioral manifold, while the Isolation Forest identifies “few and different” deviations in user-specific metrics. The experimental evaluation demonstrates that the integration of these dual signals provides a superior separation between nominal activities and behavioral anomalies compared to standalone implementations.

The resulting joint anomaly space, characterized by the hybrid index A_i , provides a mathematically consistent metric for automated security auditing and real-time session verification. The established tiered decision logic utilizing a Caution Threshold ($T_{95} = 0.349$) and a Lockout Threshold ($T_{99} = 0.463$) enables a granular response strategy that maintains authentication persistence while providing verifiable threat mitigation. This framework demonstrates high computational efficiency and adaptability, offering a scalable approach to distinguishing between systemic pattern shifts and

individual credential compromises in high throughput environments.

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